

Keysight 33512B Stimulation Control System

Operator Documentation and Quick-Reference Manual

Document scope

- Internal lab documentation for operating the graphical control interface for the Keysight 33512B function generator.
- Covers Continuous Mode, Burst Mode, Trial Mode, Digitimer D188 routing, DS5 scaling, logging, and troubleshooting.
- This document does not replace the approved study protocol, device manuals, or local safety procedures.

Critical pre-use note

- Always verify DS5 front-panel settings against the software settings before applying stimulation.
- Start at low current and increase gradually according to the study protocol.
- Verify critical timing and voltage outputs on physical hardware where possible; do not rely on the software display alone.

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1. System overview

This system provides a graphical interface for controlling a Keysight 33512B function generator in stimulation experiments. All operating modes ultimately configure the Keysight 33512B output channels, but each mode is intended for a different lab workflow.

Mode	Best used for	Typical outputs / actions
Continuous Mode	Manual setup, hardware checks, steady/repeating outputs, DS5 current scaling checks, or continuous pulse trains.	Configures one or both Keysight channels continuously.
Burst Mode	Repeated, time-locked stimulation events, paired-pulse timing, TMS triggers, synchronized stimulator triggering, and jittered schedules.	Configures one or both channels, enables burst behavior, and triggers short bursts at defined intervals.
Trial Mode	Structured participant sessions with randomized conditions, intensity changes, body-map reports, and experiment logs.	Applies trial-specific stimulation settings, routes D188 channels, logs current changes and participant reports.

The system can also communicate with a Digitimer D188 channel selector. In this setup, the Keysight 33512B provides the waveform or TTL trigger signal, while the D188 routes stimulation to a selected output channel.

DS5 scaling relationship

- When using DS5 stimulation, the software calculates the voltage sent by the function generator from the desired output current, DS5 input voltage, and DS5 output current range.
- Formula: $\text{output voltage} = \text{DS5 input voltage} \times \text{desired current} / \text{DS5 output current range}$.

2. Required pre-run hardware checks

Complete these checks before running any stimulation program.

Hardware and connection checklist

- ☐ Keysight 33512B is connected and visible to Keysight IO Libraries Suite.
- ☐ The software resource name "33512B" resolves to the correct instrument.
- ☐ Digitimer D188 driver and hardware are connected if D188 routing is being used.
- ☐ DS5 input voltage in software matches the physical DS5 front-panel setting.
- ☐ DS5 output current range in software matches the physical DS5 front-panel setting.
- ☐ Cables are connected to the expected Keysight output channel and D188 channel.
- ☐ Output is checked with low current settings before participant-facing stimulation.
- ☐ Critical settings are verified on physical hardware where possible.

3. Continuous Mode

Purpose

Continuous Mode manually configures one or both Keysight 33512B output channels. It is the simplest way to produce a waveform continuously at a selected frequency, amplitude, and pulse width.

Common use	Why it matters
Function-generator response test	Confirms the instrument is communicating before a protocol run.
Oscilloscope output check	Verifies voltage levels, pulse width, and timing.
Manual DS5 pulse train	Allows controlled low-current setup before a structured trial.
TTL trigger train	Provides an external trigger output from one channel.
Polarity comparison	Allows normal versus reversed polarity checks.
D188 channel check	Confirms selected routing before trial execution.

Main controls

Control	Function
Channel On/Off	Turns the selected channel output on or off when settings are applied.
TTL Trigger	Preconfigures the selected channel as a TTL pulse output: Pulse waveform, 5 V high, 0 V low, 1 ms pulse width. Other waveform controls are disabled while TTL Trigger is selected.
Frequency	Sets output frequency in Hz. For example, 30 Hz produces 30 pulses per second when the waveform is Pulse.
Control type	DS5 control calculates function-generator voltage from desired current and DS5 settings. Direct function-generator control allows the user to set low and high voltage directly.
Waveform	Available waveform types include Sine, Square, Pulse, and Custom.
Pulse Width	Controls pulse width in ms for Pulse waveforms.
Charge Balance	Enables or disables charge-balancing behavior for pulse-shaped stimulation waveforms.
Polarity	Normal or reversed polarity. Reversing polarity swaps the voltage direction applied by the generated waveform.
D188 Channel	Selects the D188 output channel used when D188 control is active.

Applying settings

Press the Apply button in Continuous Mode to apply the current settings. The controller applies both channel 1 and channel 2 settings.

The software uses a stateful function-generator controller. It remembers previous settings separately for each channel and only sends the required changes. If only voltage has changed, it can use a faster voltage-adjust path. If waveform, frequency, pulse width, or other settings change, it reconfigures the channel more fully.

Example workflows

Example	Goal	Suggested setup
Basic DS5 pulse train	Drive the DS5 from channel 1 with a 30 Hz pulse train.	Channel 1 On; TTL Off; DS5 control; DS5 input/output range matched to hardware; low starting current such as 1 mA; Pulse waveform; 30 Hz; 1 ms pulse width; correct D188 channel; apply and verify hardware output.
TTL trigger train	Use channel 2 as a TTL trigger output for another device.	Channel 2 On; TTL Trigger On; desired trigger frequency, for example 1 Hz; apply settings. Output should be 0-5 V with 1 ms pulse width.
Manual two-channel testing	Use channel 1 for DS5 stimulation and channel 2 as a TTL monitor or external trigger.	Configure channel 1 as DS5 pulse train; configure channel 2 with TTL Trigger On; match frequencies if they should repeat together; apply and verify both channels on an oscilloscope.

4. Burst Mode

Purpose

Burst Mode is used for discrete stimulation events rather than continuously running output. It configures the Keysight channels, enables burst behavior, and triggers bursts according to the settings chosen in the UI.

Burst Mode is useful when exact timing between two channels matters. For example, channel 1 can produce a TTL trigger for TMS while channel 2 drives or triggers another stimulator after a defined delay.

Main timing controls

Timing control	Definition / behavior
Number of stims	Total number of stimulation events in the run.
Interstim delay or stim frequency	Controls spacing between stimulation events. In delay mode, the value is seconds between events. In frequency mode, the value is Hz; for example, 0.2 Hz means one event every 5 seconds.
Pulses per stim	Number of pulses in each stimulation burst. A value of 1 gives one pulse per event. Higher values produce multi-pulse bursts.
Interpulse delay or frequency	Controls spacing between pulses within an event when multiple pulses are used. In delay mode, the value is in ms.
Channel 2 enabled/disabled	Controls whether channel 2 participates in the burst.
Channel 2 delay	Delay between channel 1 and channel 2. Positive delays place channel 2 after channel 1. Negative delays place channel 2 first and channel 1 after it.
Jitter	Adds random variation around the interstim delay. The jitter amount must be less than or equal to the interstim delay.
Random frequency	Randomizes output frequency once per burst between selected lower and upper limits. Frequency is not randomized within a burst.

Timing precision for multi-pulse bursts

- Each burst is started by a software trigger from the control computer, so the start time of each burst may have small timing jitter. A practical expected value is approximately 60 μ s.
- Once a burst has been triggered, timing inside that burst is controlled by the Keysight 33512B hardware.
- Jitter settings affect timing between stimulation events, not pulse spacing inside an already-triggered burst.

Burst Mode channel settings

Setting	Behavior
TTL Trigger On	Channel is fixed as Pulse waveform, 5 V high, 0 V low, 1 ms pulse width.
TTL Trigger Off	Channel uses its waveform, voltage/current, pulse width, charge balance, and polarity controls.
DS5 control	Software calculates voltage needed to achieve the desired DS5 current. DS5 input voltage and output current range must match physical settings.
Function-generator control	User directly sets low voltage, high voltage, waveform, and pulse width.
Charge balance	Enables charge-balanced pulse shaping when appropriate.
Polarity	Normal or reversed.
Custom waveform	Allows use of custom pulse-shaping parameters where implemented.

Running Burst Mode

1. Configure timing settings.
2. Configure channel 1.
3. Configure channel 2 if needed.
4. Choose TTL Trigger On or Off for each channel.
5. Press the burst apply/run button.
6. The system resets function-generator state, configures both channels, and starts the burst run in a background worker.
7. Use the Stop button to request that the run stop.

The burst worker stops cooperatively. It checks for stop requests before setup, after channel setup, and during the stimulation loop.

Example workflows

Example	Goal	Suggested setup / expected behavior
TMS trigger on channel 1; stimulator output on channel 2	Trigger TMS from channel 1 and drive a stimulator from channel 2 with a controlled delay.	50 events; one event every 5 s; channel 2 delay 10 ms; channel 1 TTL Trigger On; channel 2 TTL Trigger Off; channel 2 DS5 settings matched to hardware; confirm cabling; run Burst Mode. Each event sends a 0-5 V, 1 ms TTL pulse from channel 1, then channel 2 output 10 ms later.
Paired-pulse stimulation	Deliver two stimulator pulses with a fixed delay.	Set number of paired events; set interstim delay between paired events; pulses per stim = 1; enable channel 2; set channel 2 delay to paired-pulse interval such as 5 ms; configure channel waveforms/current settings; run.
Jittered TMS trigger train	Send TTL triggers with random timing variation.	Channel 1 TTL Trigger On; channel 2 disabled if not needed; set number of triggers; set mean interstim delay; turn jitter On; set jitter rate. Example: 5 s delay and 0.5 s jitter gives intervals around 4.5-5.5 s.
Multi-pulse burst	Send multiple pulses per stimulation event.	Set number of burst events; set pulses per stim; set interpulse delay; configure channel 1/channel 2 as needed; run. Burst start is software-triggered, but within-burst pulse spacing is hardware-controlled.

5. Trial Mode

Purpose

Trial Mode is used for structured stimulation experiments with participants. It helps lab members run repeatable stimulation sessions while recording participant information, randomized trial conditions, display status, electrode configuration, waveform condition, polarity, D188 channel, current intensity changes, participant reports, and timing.

Trial Mode behavior

- Trial Mode is more opinionated than Continuous Mode or Burst Mode.
- It currently applies a 30 Hz stimulation frequency.
- It uses the selected trial waveform condition to choose between Active and Sham waveform configurations.

Trial setup controls

Setup area	Required information / behavior
Participant Information	Enter participant first name, last name, and participant ID. Participant ID is required for clear file naming and later review.
Trial Conditions Table	Each row describes one condition: Electrode Config, Waveform, Polarity, and D188 Channel. Configure only the rows needed for the experiment.
DS5 Input Voltage	Must match the physical DS5 input setting.
DS5 Output Current	Must match the physical DS5 output current range.
Initial Current	Starting current for each trial.
Current Increment	Step size used by intensity increase/decrease controls.

Review dialog

Before a trial starts, the software shows a review dialog summarizing trial conditions, DS5 input voltage, DS5 output current, initial current, and current increment. Use this as the final software-side check before beginning, especially to confirm that DS5 software settings match the DS5 hardware.

How Trial Mode runs

1. Configure participant information and trial conditions.
2. Start the trial.
3. Review and confirm settings.
4. Randomize trial order.
5. Load the first trial.
6. Reset current intensity to the starting current for that trial.
7. Apply current trial settings to the Keysight/D188 hardware.
8. Increase or decrease intensity as needed.
9. Log participant reports using the body-map dialog.
10. Advance to the next configuration.
11. At the end, respond to the save prompt and save the experiment log.

Trial waveform behavior

Condition / setting	Behavior
Active waveform	Applies a pulse waveform with charge balance enabled. Pulse width is set to 1 ms. Output voltage is calculated from current intensity and DS5 settings.
Sham waveform	Applies a custom pulse-style waveform intended for sham stimulation behavior. It uses a shorter pulse width and custom pulse-shaping parameters.
Polarity	Normal or Reversed polarity is applied based on the trial condition.
D188 channel	Selected D188 channel is activated for the condition.
Frequency	Trial Mode currently uses 30 Hz.

Intensity changes

- Updates displayed current.
- Recalculates function-generator output voltage using DS5 settings.
- Applies updated settings to the hardware.
- Logs the intensity change event.

Participant logging

The Log button opens the body log dialog, which lets the user mark body locations and record participant responses. Each log is attached to the current trial entry.

Logged participant report fields
Current intensity in mA
Body-map annotations
Sensation quality
Sensation intensity
Temporal quality
Bilateral status
Motor threshold reached
Motor intensity
Motor quality

Motor temporal quality
Additional notes

Saving results

At the end of the experiment, the software asks whether to save the experiment log. Saved outputs are placed in the Trial_Logs folder.

Saved output	Purpose
JSON experiment log	Detailed trial, intensity, and participant report information.
CSV experiment summary	Quick review of trial-level results.

Trial Mode also uses an auto-backup journal during active experiments to reduce data loss if the program closes unexpectedly. When the final log is saved successfully, the backup is deleted.

Example workflows

Example	Goal	Suggested workflow
Threshold-finding session	Find participant response threshold across several electrode configurations.	Enter participant information; add trial rows; select Active waveform; set polarity and D188 channel; confirm DS5 settings; set low initial current and appropriate increment; start trial; increase gradually; log sensation/motor responses; save at end.
Active versus sham comparison	Compare participant reports across active and sham waveform conditions.	Configure trial rows with Active and Sham waveform settings; use matched electrode configurations where appropriate; confirm rows in review dialog; start randomized sequence; avoid revealing underlying condition; log reports; save final log.
D188 channel mapping check	Confirm trial rows select intended D188 outputs before a participant session.	Before participant connection, configure intended D188 channels; use low or safe test output; start sequence; step through conditions; confirm D188 indicator and physical routing; abort/finish and discard test log if not needed.

6. Recommended lab workflow

1. Use Continuous Mode first to verify hardware communication and output.
2. Use an oscilloscope or hardware monitor to confirm voltage levels and timing.
3. If timing matters, test timing in Burst Mode before participant-facing stimulation.
4. Use Trial Mode only after protocol settings have been checked.
5. Save logs immediately at the end of participant sessions.
6. Record protocol deviations in the notes/body log where appropriate.

7. Troubleshooting

Issue	First checks
Keysight does not respond	Confirm Keysight IO Libraries Suite is installed and the instrument appears under the expected resource name "33512B".
DS5 output does not match expected current	Confirm DS5 input voltage and output current range match both the software and physical DS5 front panel.
D188 channel does not change	Confirm the D188 driver is installed, the D188 is connected,

	and the selected channel is between 1 and 8.
TTL output is not visible	Confirm the correct Keysight channel is connected. TTL Trigger mode outputs a 0-5 V pulse with 1 ms pulse width.
Burst timing looks wrong	Check whether delay or frequency mode is selected. Confirm channel 2 delay sign: positive means channel 2 follows channel 1; negative means channel 2 leads channel 1.
Trial Mode will not start	Confirm that at least one trial condition is configured and the participant/trial review dialog has been confirmed.
Logs are missing	Check the Trial_Logs folder. If the program crashed, look for the auto-backup journal and ask a developer/lab maintainer before deleting anything.

8. Glossary

Term	Definition
Keysight 33512B	Two-channel function generator used to generate stimulation waveforms and TTL triggers.
DS5	Digitimer stimulator driven by a voltage signal from the function generator. The software can calculate function-generator voltage required for a desired DS5 current output.
D188	Digitimer channel selector used to route stimulation to different channels.
TTL Trigger	Digital trigger pulse. In this software, TTL Trigger mode uses 0 V low, 5 V high, and 1 ms pulse width.
Interstim delay	Time between stimulation events.
Interpulse delay	Time between pulses within an event or between channel timing events.
Charge balance	Waveform shaping intended to balance positive and negative charge over the pulse.
Polarity	Direction of the stimulation waveform. Reversed polarity swaps the voltage direction.
Sham	Control waveform or displayed condition used for experimental comparison, depending on study design.